Equal-Sum Partitions into a Fixed Number of Consecutive Group Counts

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Abstract

Fix an integer $r \geq 3$. For a positive integer m, let $f_r(m)$ be the least length of a finite indexed family of positive real numbers of total sum 1 that admits equal-sum partitions into $m, m+1, \ldots, m+r-1$ nonempty groups. Writing $d = r-1$, we prove that

$$f_r(m) = 2m + \Theta_r\Big(m^{\frac{r-2}{r-1}}\Big) = 2m + \Theta_d\Big(m^{\frac{d-1}{d}}\Big) \qquad (m \to \infty).$$

The upper bound is obtained from a d-dimensional pairing lattice: each lattice site carries two atoms, while the d coordinate directions realize the d nontrivial target pair sums exactly. A simultaneous completion lemma repairs the boundary in $O_d(m^{(d-1)/d})$ atoms. For the lower bound, a near-minimal configuration forces its weight-multiplicity function to be almost invariant under the d translations

$$h_k = \frac{1}{m} - \frac{1}{m+k} = \frac{k}{m(m+k)}, \qquad 1 \leq k \leq d.$$

These translations have no nontrivial integer relation whose coefficients lie in the box $|z_k| < (m+1)/d!$. Combining this linear-scale local arithmetic freeness with a self-contained minimal-growth argument yields a d-dimensional discrete isoperimetric inequality, which forces the excess above $2m$ to be $\Omega_d(m^{(d-1)/d})$. We also recover, with a separate sharper two-dimensional argument, the bounds

$$2m + 1 + \sqrt{\frac{m}{54}} \leq f_3(m) \leq 2m + 3\lceil\sqrt{m}\rceil$$

for the three-target problem (the upper bound for $m \geq 16$ and the lower bound for $m \geq 3$).

Keywords. Equal-sum partitions; extremal combinatorics; additive structure; translation defect; discrete isoperimetry; lattice construction.

1 Introduction and main results

A *weight system* is a finite indexed family

$$w_1, \ldots, w_N > 0, \qquad \sum_{i=1}^{N} w_i = 1.$$

The indices are the atoms of the problem: two atoms of the same numerical weight are still distinct atoms. If $q \geq 1$, we say that the system admits a *q-equal-sum partition* if its index set can be partitioned into q nonempty groups, each of total weight $1/q$.

Fix $r \geq 3$. For $m \geq 1$, define $f_r(m)$ to be the least N for which one weight system admits, possibly in different ways, q-equal-sum partitions for every

$$q \in \{m, m+1, \ldots, m+r-1\}.$$

The quantity $f_r(m)$ is finite. Indeed, subdivide $[0,1]$ simultaneously at all points $j/(m+k)$ with $0 \leq k \leq r-1$ and $0 \leq j \leq m+k$. Take the positive lengths of the common-refinement intervals as atoms. For a fixed k, group together precisely those refinement intervals lying in each interval $[j/(m+k), (j+1)/(m+k)]$; this gives $m+k$ nonempty groups of sum $1/(m+k)$. Thus the common refinement is feasible for all target counts.

Set

$$d = r-1 \geq 2, \qquad \alpha_d = \frac{d-1}{d}.$$

Our main theorem proves the fixed-r asymptotic conjecture.

Theorem 1.1 (General fixed-width asymptotic)**.** *For every fixed integer $d \geq 2$, there exist constants $M_d \in \mathbb{N}$ and $c_d > 0$ such that for all $m \geq M_d$,*

$$2m + c_d m^{\alpha_d} \leq f_{d+1}(m) \leq 2m + (2d-1)\lceil m^{1/d} \rceil^{d-1} + \frac{(d-1)(d-2)}{2}, \tag{1}$$

where one may take

$$c_d = \frac{2^{\alpha_d}}{12(d+4)(e\,d!)^{1/d}}. \tag{2}$$

Consequently, for every fixed $r \geq 3$,

$$f_r(m) = 2m + \Theta_r\left(m^{\frac{r-2}{r-1}}\right), \qquad \lim_{m\to\infty} \frac{f_r(m)}{m} = 2.$$

For complete explicitness, the proof works with

$$M_d^{\mathrm{up}} := \max\left\{2^{d(d-1)},\ 3d,\ \left\lfloor [d(d+1)]^{d/(d-1)} \right\rfloor + 1\right\}, \tag{3}$$

$$M_d^{\mathrm{iso}} := \max\left\{4d!,\ \left\lfloor \left(3e\,4^d (d!)^{d+1}\right)^{1/(d-1)} \right\rfloor + 1\right\}, \tag{4}$$

and one may take $M_d = \max\{M_d^{\mathrm{up}}, M_d^{\mathrm{iso}}\}$.

The case $d = 2$ admits a stronger lower constant and a better explicit upper threshold.

Theorem 1.2 (Three consecutive group counts)**.** *For every integer $m \geq 16$,*

$$2m + 1 + \sqrt{\frac{m}{54}} \leq f_3(m) \leq 2m + 3\lceil \sqrt{m} \rceil.$$

The lower bound holds for every $m \geq 3$.

The proofs of Theorems 1.1 and 1.2 are entirely contained below.

2 Notation and a simultaneous completion lemma

Fix $d \geq 2$ and a feasible system for the target group counts

$$m, m+1, \ldots, m+d.$$

Let P_k denote the partition into $m+k$ groups, and set

$$t_k := \frac{1}{m+k} \qquad (0 \leq k \leq d). \tag{5}$$

Thus every group of P_k has total weight t_k.

Every atom has weight at most t_d, because it belongs to a P_d-group of total weight t_d. Hence, for every $k < d$, no group of P_k can be a singleton, since $t_k > t_d$. In particular,

$$N \geq 2(m+d-1). \tag{6}$$

It is convenient to write

$$N = 2m + E. \tag{7}$$

Then $E \geq 2d-2$.

We shall repeatedly use the following completion device.

Lemma 2.1 (Simultaneous completion)**.** *Suppose a finite collection of positive-weight atoms has total mass $\sigma < 1$. For each $k = 0, \ldots, d$, suppose these atoms have already been assigned to prescribed groups of a prospective P_k so that:*

(i) *every existing atom is assigned to exactly one group in each P_k;*

(ii) *no prescribed group exceeds its target weight t_k;*

(iii) *the total target capacity of all groups of P_k is 1.*

Let g_k be the number of groups of P_k that are not yet full. Then one can add at most

$$\sum_{k=0}^{d} g_k - d \tag{8}$$

new positive-weight atoms so that every group in every P_k becomes exactly full. If all existing weights and all t_k are rational, then the new weights may be chosen rational.

Proof. Let $\rho = 1 - \sigma > 0$. Necessarily $g_k \geq 1$ for every k, since otherwise all groups of P_k would already be full and the existing atoms would have total mass 1. For a fixed k, list the positive residual capacities of the g_k unfilled groups in any order. They sum to ρ, so they determine a partition of $[0, \rho]$ into g_k consecutive positive intervals. Take the common refinement of the $d+1$ interval partitions obtained in this way. Altogether they contribute at most

$$\sum_{k=0}^{d} (g_k - 1)$$

distinct internal endpoints, so the common refinement has at most

$$1 + \sum_{k=0}^{d} (g_k - 1) = \sum_{k=0}^{d} g_k - d$$

positive subintervals. Use the lengths of these subintervals as the new atom weights. In partition P_k, assign each new atom to the unique residual-capacity interval containing the corresponding common-refinement subinterval. Every residual interval is thereby filled exactly, and each new atom is used once in each partition. If the residual capacities are rational, then every refinement endpoint, and hence every new atom weight, is rational. □

3 The upper bound: a d-dimensional pairing lattice

Throughout this section assume $m \geq M_d^{\mathrm{up}}$, with M_d^{up} defined in (3). Set

$$s := \lceil m^{1/d} \rceil, \qquad b := s^{d-1}, \qquad K := m - b + 1. \tag{9}$$

The choice $m \geq 2^{d(d-1)}$ implies $b \leq m$: indeed,

$$s \leq m^{1/d} + 1 \leq 2m^{1/d},$$

so

$$b \leq 2^{d-1} m^{(d-1)/d} \leq m.$$

Thus $K \geq 1$. Also $b \geq 2$, hence $K < m$, and $K \leq m \leq s^d$.

3.1 A compressed box

Let

$$Q := \{0, 1, \ldots, s-1\}^d.$$

For $a = (a_1, \ldots, a_d) \in Q$, define its base-$s$ index

$$\iota(a) := a_1 + a_2 s + \cdots + a_d s^{d-1}.$$

Take

$$L := \{a \in Q : \iota(a) < K\}. \tag{10}$$

Then $|L| = K$. For each coordinate direction e_k, the intersection of L with any line parallel to e_k is either empty or an initial consecutive segment of that lattice line, because after all other coordinates are fixed the condition $\iota(a) < K$ is a strict upper bound on a_k.

Let H_k be the number of nonempty e_k-parallel lattice lines meeting L. Since such a line is determined by the other $d-1$ coordinates,

$$H_k \leq s^{d-1} = b. \tag{11}$$

If a nonempty line contains ℓ points of L, it contributes $\ell - 1$ adjacent pairs in direction e_k. Summing over the H_k nonempty lines gives exactly

$$K - H_k \tag{12}$$

adjacent pairs in direction e_k.

3.2 Two atoms per lattice site

For $1 \le k \le d$, define

$$\Delta_k := t_0 - t_k = \frac{k}{m(m+k)}. \tag{13}$$

For $a = (a_1, \dots, a_d) \in L$, put

$$z_a := \sum_{k=1}^{d} a_k \Delta_k, \qquad W := \max_{a \in L} z_a. \tag{14}$$

At every $a \in L$, introduce two distinct atoms U_a, V_a of weights

$$u_a := \frac{t_0 - W}{2} + z_a, \qquad v_a := t_0 - u_a. \tag{15}$$

We verify that these are positive and, in fact, strictly smaller than t_d. Since $s - 1 < m^{1/d}$ and $\Delta_k \le k/m^2$,

$$0 \le W \le (s-1) \sum_{k=1}^{d} \Delta_k < \frac{d(d+1)}{2}\, m^{-2+1/d}. \tag{16}$$

Because $m \ge 3d$,

$$2t_d - t_0 = \frac{m-d}{m(m+d)} \ge \frac{1}{2m}.$$

The definition of M_d^{up} also gives

$$m^{(d-1)/d} > d(d+1),$$

so the right-hand side of (16) is $< 1/(2m)$. Consequently,

$$W < 2t_d - t_0. \tag{17}$$

Since $0 \le z_a \le W$,

$$u_a, v_a \in \left[\frac{t_0 - W}{2}, \frac{t_0 + W}{2}\right] \subset (0, t_d).$$

3.3 Exact pairings in all $d+1$ partitions

For every $a \in L$,

$$u_a + v_a = t_0. \tag{18}$$

Moreover, whenever $a, a + e_k \in L$,

$$z_{a+e_k} - z_a = \Delta_k,$$

and hence

$$u_a + v_{a+e_k} = t_0 - \Delta_k = t_k. \tag{19}$$

Thus P_0 uses the K disjoint pairs $\{U_a, V_a\}$. For $1 \le k \le d$, along each nonempty e_k-line, use every adjacent pair

$$\{U_a, V_{a+e_k}\}.$$

These pairs are disjoint: on a line of length ℓ, they use U at positions $0, \dots, \ell - 2$ and V at positions $1, \dots, \ell - 1$. Exactly two atoms, namely the initial V and the terminal U, remain unpaired on that line. Therefore P_k contains $K - H_k$ full binary groups and exactly $2H_k$ unpaired existing atoms.

3.4 Boundary bookkeeping and completion

In P_0, keep the K full binary groups and introduce $m - K = b - 1$ empty groups. Hence

$$g_0 = b - 1. \tag{20}$$

For $1 \le k \le d$, put each of the $2H_k$ unpaired atoms alone into a temporarily unfilled group. The number of currently occupied groups is

$$(K - H_k) + 2H_k = K + H_k \le m + 1 \le m + k,$$

using (11). We may therefore add

$$m + k - K - H_k = b - 1 + k - H_k \ge 0$$

empty groups. The total number of unfilled groups in P_k is then

$$g_k = 2H_k + b - 1 + k - H_k = b - 1 + k + H_k. \tag{21}$$

The $2K$ existing atoms have total mass $Kt_0 = K/m < 1$. Every singleton group has positive residual capacity because each existing atom has weight $< t_d \le t_k$. Lemma 2.1 therefore adds at most $\sum_{k=0}^{d} g_k - d$ atoms. The final number of atoms is at most

$$\begin{aligned} 2K + \sum_{k=0}^{d} g_k - d &= 2m + (d-1)b + \sum_{k=1}^{d} H_k + \frac{(d-1)(d-2)}{2} \\ &\le 2m + (2d-1)b + \frac{(d-1)(d-2)}{2}. \end{aligned}$$

We have proved the following proposition.

Proposition 3.1 (General upper bound)**.** *For every fixed $d \ge 2$ and every $m \ge M_d^{\mathrm{up}}$,*

$$f_{d+1}(m) \le 2m + (2d-1)\lceil m^{1/d} \rceil^{d-1} + \frac{(d-1)(d-2)}{2}.$$

The constructed weights may all be chosen rational.

4 Binary defects and approximate translation invariance

We now prove the lower bound. Fix a feasible weight system of $N = 2m + E$ atoms. For each k, let I_k denote the number of atoms lying in nonbinary groups of P_k; here "nonbinary" means a singleton or a group of size at least three.

4.1 The partitions P_k with $k < d$

For $k < d$, there are no singleton groups. Let a_k be the number of groups of P_k of size at least three. Since P_k has $m + k$ groups,

$$\sum_{G \in P_k} (|G| - 2) = N - 2(m + k) = E - 2k. \tag{22}$$

Every summand is nonnegative; in particular $E - 2k \ge 0$, and since each group counted by a_k contributes at least 1, we have $a_k \le E - 2k$. Moreover,

$$I_k = \sum_{\substack{G \in P_k \\ |G| \ge 3}} |G| = (E - 2k) + 2a_k \le 3(E - 2k) \le 3E. \tag{23}$$

4.2 Singletons in the final partition

We need a separate estimate for P_d, which may contain singletons. Construct a graph $\mathcal{H}$ on the atoms by joining the two atoms in every binary group of P_0 by a color-0 edge and the two atoms in every binary group of P_1 by a color-1 edge. Every vertex has degree at most one in each color, so each connected component has maximum degree at most two and every nontrivial component is alternating.

There are no alternating cycles. Indeed, traversing a color-k edge replaces a weight x by $t_k - x$. Thus a color-0 step followed by a color-1 step sends

$$x \longmapsto t_1 - (t_0 - x) = x - (t_0 - t_1),$$

whereas a color-1 step followed by a color-0 step sends x to $x + (t_0 - t_1)$. Any alternating cycle has even length $2q \geq 2$; after choosing an orientation, its successive pairs of colors are all $0, 1$ or all $1, 0$, so one full traversal changes the starting weight by $\mp q(t_0 - t_1) \neq 0$. This cannot return to the starting vertex. Hence $\mathcal{H}$ is a forest of paths, isolated vertices included.

The numbers of binary groups in P_0 and P_1 are $m - a_0$ and $m + 1 - a_1$, respectively. Thus the number τ of path components is

$$\begin{aligned} \tau &= N - \big[(m - a_0) + (m + 1 - a_1)\big] \\ &= E - 1 + a_0 + a_1 \leq 3E - 3. \end{aligned} \tag{24}$$

Let ℓ be the number of singleton groups of P_d. Equivalently, ℓ is the number of atoms of weight t_d.

Lemma 4.1. *Every path component of $\mathcal{H}$ contains at most two atoms of weight t_d. Consequently,*

$$\ell \leq 2\tau \leq 6E - 6. \tag{25}$$

Proof. Split the vertices of a path into its two parity classes according to graph distance from one endpoint. If two distinct vertices in the same parity class are $2q$ edges apart, then repeated application of the two-step rule above shows that their weights differ by $\pm q(t_0 - t_1) \neq 0$. Hence each parity class contains at most one vertex of weight t_d, and the whole path contains at most two. □

For P_d, summing $|G| - 2$ over all groups gives $E - 2d$. A singleton contributes -1, so

$$\sum_{\substack{G \in P_d \\ |G| \geq 3}} (|G| - 2) = E - 2d + \ell. \tag{26}$$

For every integer $q \geq 3$, $q \leq 3(q - 2)$. Hence, using (25),

$$\begin{aligned} I_d &\leq \ell + 3(E - 2d + \ell) \\ &\leq 27E. \end{aligned} \tag{27}$$

4.3 Multiplicity functions and translation defects

Define the finitely supported multiplicity function

$$\mu(x) := \#\{i : w_i = x\}, \qquad x \in \mathbb{R}. \tag{28}$$

For finitely supported functions $\nu_1, \nu_2 : \mathbb{R} \to \mathbb{R}$, define

$$V(\nu_1, \nu_2) := \frac{1}{2} \sum_{x \in \mathbb{R}} |\nu_1(x) - \nu_2(x)|. \tag{29}$$

For $h, a \in \mathbb{R}$, define

$$(T_h\nu)(x) := \nu(x-h), \qquad (R_a\nu)(x) := \nu(a-x). \tag{30}$$

Finally put

$$B_h(\mu) := V(\mu, T_h\mu). \tag{31}$$

Write $\mu = \mu_k^{\mathrm{bin}} + \eta_k$, where μ_k^{bin} is the multiplicity function of atoms lying in binary groups of P_k and η_k is that of the remaining atoms. Every binary group consists of weights x and $t_k - x$, so

$$R_{t_k}\mu_k^{\mathrm{bin}} = \mu_k^{\mathrm{bin}}.$$

Also $\sum_x \eta_k(x) = I_k$. Therefore

$$\begin{aligned} V(\mu, R_{t_k}\mu) &= V(\eta_k, R_{t_k}\eta_k) \\ &\le \frac{1}{2}\left(\sum_x \eta_k(x) + \sum_x (R_{t_k}\eta_k)(x)\right) = I_k. \end{aligned} \tag{32}$$

Also $R_aR_b = T_{a-b}$. Since reflections preserve V, the triangle inequality yields

$$B_{t_0-t_k}(\mu) \le V(\mu, R_{t_0}\mu) + V(\mu, R_{t_k}\mu) \le I_0 + I_k. \tag{33}$$

Define

$$h_k := t_0 - t_k = \frac{k}{m(m+k)}, \qquad 1 \le k \le d. \tag{34}$$

By (23), (27), and (33),

$$\sum_{k=1}^{d} B_{h_k}(\mu) \le 6(d+4)E. \tag{35}$$

Indeed, $B_{h_k}(\mu) \le 6E$ for $1 \le k < d$, while $B_{h_d}(\mu) \le 30E$.

5 Local arithmetic freeness of the translation steps

The special translations (34) are globally rationally dependent, but inside the coefficient box $|z_k| < (m+1)/d!$ they behave like d independent generators. This linear-size relation-free window is all that the lower-bound argument needs.

Lemma 5.1 (No short integer relation)**.** *Let $z_1, \dots, z_d \in \mathbb{Z}$. If*

$$\sum_{k=1}^{d} z_k h_k = 0 \tag{36}$$

and

$$|z_k| d! < m+1 \qquad (1 \le k \le d), \tag{37}$$

then $z_1 = \cdots = z_d = 0$.

Proof. Using (34), cancel the common factor $1/m$ in (36) to obtain

$$\sum_{k=1}^{d} \frac{z_k k}{m+k} = 0.$$

Multiply by

$$P := \prod_{j=1}^{d}(m+j).$$

For a fixed k, reduce the resulting integer identity modulo $m+k$. Every term except the kth is divisible by $m+k$, so

$$m+k \mid z_k k \prod_{j\neq k}(m+j).$$

Modulo $m+k$, the product on the right is congruent to

$$z_k k \prod_{j\neq k}(j-k).$$

Its absolute value is

$$|z_k|\, k!(d-k)! \leq |z_k| d! < m+1 \leq m+k.$$

The only integer of absolute value $< m+k$ divisible by $m+k$ is 0. Since $k\prod_{j\neq k}(j-k) \neq 0$, we get $z_k = 0$. This holds for every k. □

Let

$$S := \{0, h_1, \ldots, h_d\}. \tag{38}$$

For a positive integer n, write nS for the n-fold sumset $S + \cdots + S$.

Corollary 5.2 (Simplex growth)**.** *If*

$$n \leq \left\lfloor \frac{m}{2d!} \right\rfloor, \tag{39}$$

then all sums

$$\sum_{k=1}^{d} \alpha_k h_k, \qquad \alpha_k \in \mathbb{Z}_{\geq 0}, \quad \sum_{k=1}^{d} \alpha_k \leq n,$$

are distinct. Consequently,

$$|nS| = \binom{n+d}{d} \geq \frac{n^d}{d!}. \tag{40}$$

Proof. If two displayed sums were equal, their coefficient difference z_k would satisfy $|z_k| \leq n$, and hence

$$|z_k| d! \leq nd! \leq \frac{m}{2} < m+1.$$

Lemma 5.1 would force all coefficient differences to vanish. The number of d-tuples of nonnegative integers with total at most n is $\binom{n+d}{d}$. Finally,

$$\binom{n+d}{d} = \prod_{j=1}^{d} \frac{n+j}{j} \geq \frac{n^d}{d!}.$$

□

6 A minimal-growth lemma and discrete isoperimetry

For a finite set $F \subset \mathbb{R}$ and $h \neq 0$, define

$$c_h(F) := |F \setminus (F+h)|. \tag{41}$$

Since $|F| = |F+h|$,

$$c_h(F) = |(F+h) \setminus F| = \frac{1}{2} \sum_{x \in \mathbb{R}} |\mathbf{1}_F(x) - \mathbf{1}_F(x-h)|. \tag{42}$$

Equivalently, if points of F differing by h are connected, $c_h(F)$ is the number of resulting finite h-chains.

The next lemma is the only sumset-growth input we need, and we include its proof.

Lemma 6.1 (Minimal-growth iteration)**.** *Let A, B be finite nonempty subsets of an abelian group. Choose a nonempty $X \subseteq A$ for which*

$$K := \frac{|X+B|}{|X|}$$

is minimal among all nonempty subsets of A. Then for every finite set C,

$$|X+B+C| \leq K|X+C|. \tag{43}$$

Consequently, for every integer $n \geq 1$,

$$|X+nB| \leq K^n|X|. \tag{44}$$

Proof. We prove (43) by induction on $|C|$. The empty case is trivial, and the one-point case is just translation invariance of cardinality. Suppose the assertion holds for a finite set C' and let $c \notin C'$, $C = C' \cup \{c\}$. Define

$$D := \{x \in X : x + c \in X + C'\}.$$

Then

$$|X+C| = |X+C'| + |X| - |D|. \tag{45}$$

Moreover,

$$D + B + c \subseteq X + B + C'.$$

Hence the number of new elements contributed by $X+B+c$ beyond $X+B+C'$ is at most

$$|X+B| - |D+B|.$$

If $D \neq \varnothing$, the minimality of X gives $|D+B| \geq K|D|$; if $D = \varnothing$, the same inequality is trivially true. Therefore, using the induction hypothesis and (45),

$$\begin{aligned} |X+B+C| &\leq |X+B+C'| + K(|X| - |D|) \\ &\leq K|X+C'| + K(|X| - |D|) \\ &= K|X+C|. \end{aligned}$$

This proves (43). Applying it successively with $C = (n-1)B, (n-2)B, \ldots, B$ gives (44). □

We can now prove the required isoperimetric estimate.

Lemma 6.2 (Local d-dimensional isoperimetry)**.** *Fix $d \geq 2$, and assume $m \geq M_d^{\mathrm{iso}}$, where M_d^{iso} is defined in* (4)*. Let h_k be as in* (34)*. Then every nonempty finite $F \subset \mathbb{R}$ with $|F| \leq 3m$ satisfies*

$$\sum_{k=1}^{d} c_{h_k}(F) \geq \kappa_d |F|^{(d-1)/d}, \qquad \kappa_d := \frac{1}{2(e\, d!)^{1/d}}. \tag{46}$$

Proof. Write

$$M := |F|, \qquad C := \sum_{k=1}^{d} c_{h_k}(F).$$

Each h_k is nonzero and F is finite and nonempty, so $C > 0$. By (38) and (42),

$$|F + S| \leq M + C. \tag{47}$$

Apply Lemma 6.1 with $A = F$ and $B = S$, and choose the minimizing nonempty $X \subseteq F$. Then

$$K = \frac{|X + S|}{|X|} \leq \frac{|F + S|}{|F|} \leq 1 + \frac{C}{M}. \tag{48}$$

If $C \geq M/2$, then, since $M \geq 1$ and $\kappa_d \leq 1/2$,

$$C \geq \frac{1}{2} M \geq \kappa_d M^{(d-1)/d},$$

and we are done. Assume henceforth that $C < M/2$, so $M/C > 2$.

Put

$$L := \left\lfloor \frac{m}{2d!} \right\rfloor, \qquad n := \min\left\{ \left\lfloor \frac{M}{C} \right\rfloor, L \right\}. \tag{49}$$

Since $C < M/2$, we have $M/C > 2$; moreover $m \geq 4d!$ gives $L \geq 1$, so $n \geq 1$. Also $n \leq M/C$, and hence $nC/M \leq 1$. Lemma 6.1, (48), and $1 + x \leq e^x$ give

$$|X + nS| \leq K^n |X| \leq \exp\left(\frac{nC}{M}\right) M \leq eM. \tag{50}$$

Since X is nonempty, any fixed $x_0 \in X$ gives $x_0 + nS \subseteq X + nS$, and therefore

$$|X + nS| \geq |nS|. \tag{51}$$

We claim that $n \neq L$. Because $m \geq 4d!$,

$$L \geq \frac{m}{4d!}.$$

If $n = L$, Corollary 5.2, (50), and $M \leq 3m$ would imply

$$\frac{m^d}{4^d (d!)^{d+1}} \leq \frac{L^d}{d!} \leq |nS| \leq eM \leq 3em.$$

But the definition of M_d^{iso} makes the first and last expressions satisfy the strict reverse inequality, namely

$$m^{d-1} > 3e\, 4^d (d!)^{d+1}.$$

This contradiction proves $n \neq L$. Hence

$$n = \left\lfloor \frac{M}{C} \right\rfloor.$$

Since $M/C > 2$,

$$n \geq \frac{M}{2C}. \tag{52}$$

Combining Corollary 5.2, (50), (51), and (52), we obtain

$$\frac{1}{d!}\left(\frac{M}{2C}\right)^d \leq \frac{n^d}{d!} \leq |nS| \leq eM.$$

Rearranging gives

$$C \geq \frac{1}{2(e\,d!)^{1/d}} M^{(d-1)/d},$$

as required. □

7 The general lower bound

We lift Lemma 6.2 from sets to the multiplicity function μ. For every integer $q \geq 1$, define the level set

$$F_q := \{x \in \mathbb{R} : \mu(x) \geq q\}. \tag{53}$$

Only finitely many F_q are nonempty, and

$$\sum_{q \geq 1} |F_q| = \sum_x \mu(x) = N. \tag{54}$$

The pointwise identity

$$|u - v| = \sum_{q \geq 1} \left| \mathbf{1}_{\{u \geq q\}} - \mathbf{1}_{\{v \geq q\}} \right| \qquad (u, v \in \mathbb{Z}_{\geq 0})$$

and (42) imply

$$B_h(\mu) = \sum_{q \geq 1} c_h(F_q). \tag{55}$$

Proposition 7.1 (General lower bound)**.** *For every fixed $d \geq 2$ and every $m \geq M_d^{\mathrm{iso}}$,*

$$f_{d+1}(m) \geq 2m + c_d m^{(d-1)/d},$$

where c_d is given by (2)*.*

Proof. Let a feasible system have $N = 2m + E$ atoms. If $E \geq m$, then, because $0 < c_d < 1$ and $m \geq 1$,

$$E \geq m \geq c_d m^{(d-1)/d},$$

so there is nothing to prove. Assume $E < m$. Then

$$N = 2m + E < 3m,$$

and every nonempty level set F_q satisfies $|F_q| \le N < 3m$. Lemma 6.2 and (55) give

$$\sum_{k=1}^{d} B_{h_k}(\mu) = \sum_{q\ge 1}\sum_{k=1}^{d} c_{h_k}(F_q) \ge \kappa_d \sum_{q\ge 1} |F_q|^{\alpha_d}. \tag{56}$$

For $0 < \alpha_d < 1$, the function $x \mapsto x^{\alpha_d}$ is subadditive on $[0,\infty)$, so

$$\sum_{q\ge 1} |F_q|^{\alpha_d} \ge \left(\sum_{q\ge 1} |F_q|\right)^{\alpha_d} = N^{\alpha_d}. \tag{57}$$

Combining (35), (56), and (57),

$$6(d+4)E \ge \kappa_d N^{\alpha_d} \ge \kappa_d (2m)^{\alpha_d}.$$

Hence

$$E \ge \frac{2^{\alpha_d}\kappa_d}{6(d+4)} m^{\alpha_d} = c_d m^{\alpha_d},$$

which proves the proposition. □

Propositions 3.1 and 7.1, with $M_d = \max\{M_d^{\mathrm{up}}, M_d^{\mathrm{iso}}\}$, prove the two-sided estimate in Theorem 1.1. Since $\lceil m^{1/d} \rceil \le 2m^{1/d}$ for $m \ge 1$, the upper error term is $O_d(m^{(d-1)/d})$; dividing both bounds by m also gives $f_{d+1}(m)/m \to 2$. Thus all asserted consequences of Theorem 1.1 follow.

8 Sharper bounds for three consecutive group counts

We now prove Theorem 1.2. This section is logically independent of the general lower-bound machinery in Sections 5–7 except for the basic defect notation, and it gives a substantially better constant when $d = 2$.

Write P_0, P_1, P_2 for the partitions into $m, m+1, m+2$ groups, and put

$$t_0 = \frac{1}{m}, \qquad t_1 = \frac{1}{m+1}, \qquad t_2 = \frac{1}{m+2}.$$

Again write $N = 2m + E$.

8.1 The upper bound for $m \ge 16$

Set $s = \lceil \sqrt{m} \rceil$ and $K = m - s + 1$. For $m \ge 16$ we have $2 \le s \le m$, hence $1 \le K < m$, and also $K \le m \le s^2$; thus all purely combinatorial parts of the two-dimensional construction in Section 3 apply. Use that construction with

$$\Delta_1 = t_0 - t_1, \qquad \Delta_2 = t_0 - t_2.$$

For $d = 2$, the quantity W in (14) satisfies the sharper estimate

$$W \le (s-1)(\Delta_1 + \Delta_2) = (s-1)\frac{3m+4}{m(m+1)(m+2)}. \tag{58}$$

For $m \geq 16$, $s - 1 < \sqrt{m} \leq m/4$, and

$$(m-2)(m+1) - \frac{m(3m+4)}{4} = \frac{m(m-8)}{4} - 2 > 0.$$

Therefore (58) gives

$$W < \frac{(m-2)(m+1)}{m(m+1)(m+2)} = 2t_2 - t_0,$$

so all constructed atoms lie in $(0, t_2)$ exactly as in Section 3. The boundary formula of Proposition 3.1 specializes to

$$f_3(m) \leq 2m + 3s = 2m + 3\lceil\sqrt{m}\rceil.$$

8.2 Sharper defect bounds

For $k = 0, 1$, let a_k be the number of groups of size at least three in P_k, and let I_k be the number of atoms in nonbinary groups. From (22),

$$I_0 \leq 3E, \qquad I_1 \leq 3E - 6. \tag{59}$$

The alternating P_0–P_1 graph from Section 4 is a path forest with

$$\tau = E - 1 + a_0 + a_1 \leq 3E - 3 \tag{60}$$

components.

Lemma 8.1. *Assume $m \geq 3$. Every component of the alternating P_0–P_1 graph contains at most one atom of weight t_2. Hence the number ℓ of singleton groups of P_2 satisfies*

$$\ell \leq 3E - 3. \tag{61}$$

Proof. Two vertices at a positive even distance $2q$ have weights differing by $\pm q(t_0 - t_1)$, so they cannot both have weight t_2.

Suppose instead that two vertices of weight t_2 are at odd distance. An odd alternating composition of the reflections $x \mapsto t_0 - x$ and $x \mapsto t_1 - x$ has the form

$$x \longmapsto t_0 + j(t_0 - t_1) - x$$

for some $j \in \mathbb{Z}$: this follows immediately by pairing successive reflections, each pair being a translation by $\pm(t_0 - t_1)$, with one final reflection left over. Thus

$$2t_2 = t_0 + j(t_0 - t_1). \tag{62}$$

Let

$$D := m(m+1)(m+2).$$

Multiplying (62) by D gives

$$2m(m+1) = (m+1)(m+2) + j(m+2).$$

The right-hand side is divisible by $m + 2$, whereas

$$2m(m+1) \equiv 4 \pmod{m+2}.$$

Since $m + 2 \geq 5$, this is impossible. Therefore each path contains at most one t_2-atom. Equation (61) follows from (60). □

For P_2,

$$\sum_{\substack{G\in P_2\\ |G|\geq 3}} (|G|-2) = E - 4 + \ell.$$

Using $q \leq 3(q-2)$ for $q \geq 3$ and Lemma 8.1,

$$I_2 \leq \ell + 3(E - 4 + \ell) \leq 15E - 24. \tag{63}$$

Define

$$\delta_{01} := t_0 - t_1 = \frac{1}{m(m+1)}, \qquad \delta_{12} := t_1 - t_2 = \frac{1}{(m+1)(m+2)}. \tag{64}$$

The reflection argument (33), now applied to (P_0, P_1) and (P_1, P_2), gives

$$B_{\delta_{01}}(\mu) \leq I_0 + I_1 \leq 6E - 6, \tag{65}$$

$$B_{\delta_{12}}(\mu) \leq I_1 + I_2 \leq 18E - 30. \tag{66}$$

8.3 A two-chain intersection inequality

Lemma 8.2 (Chain-intersection bound)**.** *Let p, q be coprime positive integers, let $\varepsilon > 0$, and let $\mu : \mathbb{R} \to \mathbb{Z}_{\geq 0}$ be finitely supported. If*

$$B_{p\varepsilon}(\mu) < p,$$

then

$$\sum_x \mu(x) \leq B_{p\varepsilon}(\mu) B_{q\varepsilon}(\mu). \tag{67}$$

Proof. We first prove the set version. For finite $F \subset \mathbb{R}$, $c_h(F)$ is the number of h-chains in F. Assume

$$c_{p\varepsilon}(F) < p.$$

Any $p\varepsilon$-chain and any $q\varepsilon$-chain meet in at most one point. Indeed, if $x < y$ were two common points, then

$$y - x = ap\varepsilon = bq\varepsilon$$

for positive integers a, b. Since $\gcd(p, q) = 1$, $y - x$ is a positive multiple of $pq\varepsilon$. The common $q\varepsilon$-chain therefore contains

$$x, x + q\varepsilon, \ldots, x + (p-1)q\varepsilon.$$

These p points lie in pairwise distinct $p\varepsilon$-chains: if the points with indices $i < j$ lay in the same such chain, then $(j-i)q\varepsilon$ would be an integer multiple of $p\varepsilon$; since $\gcd(p, q) = 1$, this would force $p \mid (j - i)$, impossible for $1 \leq j - i \leq p - 1$. This contradicts $c_{p\varepsilon}(F) < p$.

Thus each point of F is uniquely determined by the ordered pair consisting of its $p\varepsilon$-chain and its $q\varepsilon$-chain. Hence

$$|F| \leq c_{p\varepsilon}(F) c_{q\varepsilon}(F). \tag{68}$$

Now define the level sets $F_j = \{x : \mu(x) \geq j\}$. By (54) and (55),

$$\sum_x \mu(x) = \sum_{j\geq 1} |F_j|, \qquad B_h(\mu) = \sum_{j\geq 1} c_h(F_j).$$

The hypothesis $B_{p\varepsilon}(\mu) < p$ implies $c_{p\varepsilon}(F_j) < p$ for every nonempty level set. Applying (68) level by level and then using nonnegativity,

$$\begin{aligned}\sum_x \mu(x) &\le \sum_{j\ge 1} c_{p\varepsilon}(F_j)c_{q\varepsilon}(F_j)\\ &\le \left(\sum_{j\ge 1} c_{p\varepsilon}(F_j)\right)\left(\sum_{j\ge 1} c_{q\varepsilon}(F_j)\right)\\ &= B_{p\varepsilon}(\mu)B_{q\varepsilon}(\mu).\end{aligned}$$

□

8.4 Proof of the sharp two-dimensional lower bound

Let

$$g := \gcd(m,2), \qquad p := \frac{m+2}{g}, \qquad q := \frac{m}{g}, \qquad \varepsilon := \frac{g}{m(m+1)(m+2)}. \tag{69}$$

Then $\gcd(p,q) = 1$ and

$$\delta_{01} = p\varepsilon, \qquad \delta_{12} = q\varepsilon.$$

We split into two cases.

If

$$6(E-1) < p, \tag{70}$$

then (65) gives $B_{p\varepsilon}(\mu) < p$. Lemma 8.2 and (65)–(66) imply

$$N \le (6E-6)(18E-30) \le 108(E-1)^2.$$

Since $N \ge 2m$,

$$E \ge 1 + \sqrt{\frac{m}{54}}. \tag{71}$$

If instead $6(E-1) \ge p$, then $g \le 2$ gives

$$E - 1 \ge \frac{p}{6} = \frac{m+2}{6g} \ge \frac{m+2}{12} \ge \frac{m}{12}.$$

For $m \ge 3$,

$$\frac{m}{12} \ge \sqrt{\frac{m}{54}},$$

so (71) again holds. Therefore every feasible system with $m \ge 3$ satisfies

$$N = 2m + E \ge 2m + 1 + \sqrt{\frac{m}{54}}.$$

Together with the upper bound already proved for $m \ge 16$, this completes the proof of Theorem 1.2.

9 Structural interpretation and further questions

The exponent $(d-1)/d$ arises from the same geometry on both sides of the argument. In the construction, the d distinct target differences $t_0 - t_k$ are assigned to the d coordinate directions of a lattice box; its boundary has order $m^{(d-1)/d}$. In the lower bound, any system with excess E has only $O_d(E)$ total defect under those same d translations. Although the translations are globally rationally dependent, Lemma 5.1 shows that they have no nontrivial integer relation with $|z_k| < (m+1)/d!$; hence Corollary 5.2 gives genuine d-dimensional simplex growth throughout the range $n \le m/(2d!)$. The minimal-growth argument then turns that local growth into a discrete isoperimetric lower bound of the same exponent.

Several natural problems remain.

1. Determine the optimal constants in the second-order term. In particular, decide whether

$$\frac{f_r(m) - 2m}{m^{(r-2)/(r-1)}}$$

converges for fixed $r \ge 3$.

2. Prove a stability theorem describing all configurations with

$$f_r(m) = 2m + O_r\left(m^{(r-2)/(r-1)}\right).$$

The upper construction suggests a compressed lattice description of near-extremizers.

3. Study the regime $r = r(m) \to \infty$. The present method is not uniform in r, because its local-freeness radius deteriorates through the factor $d!$.